

Nonlinear Dynamics: Coupled Oscillators in Fly Walking



Klaus Wimmer

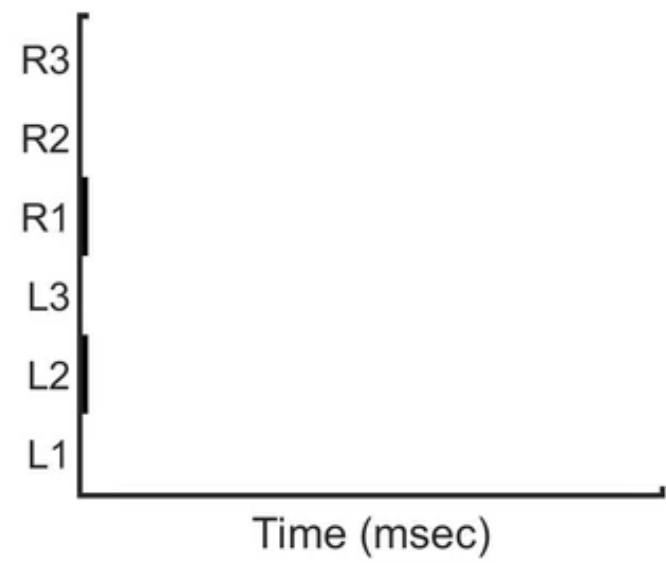
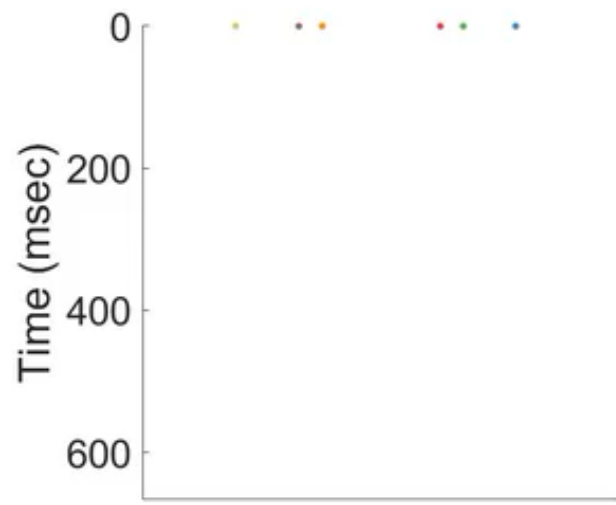
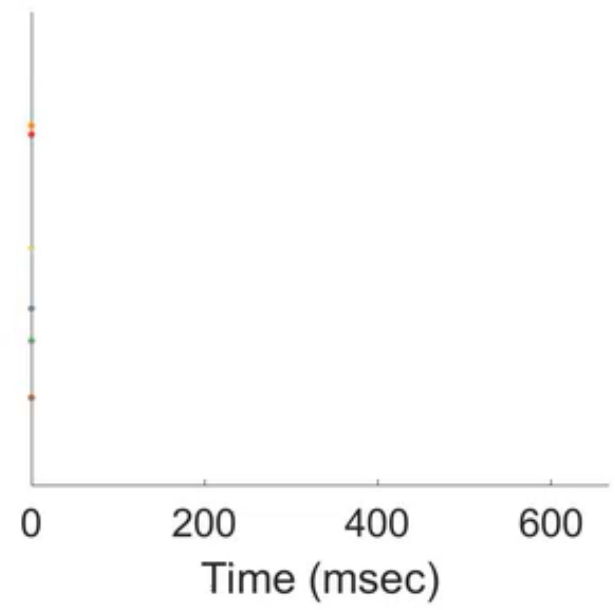
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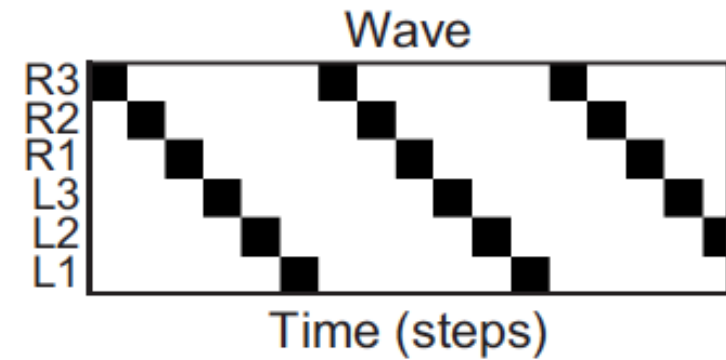
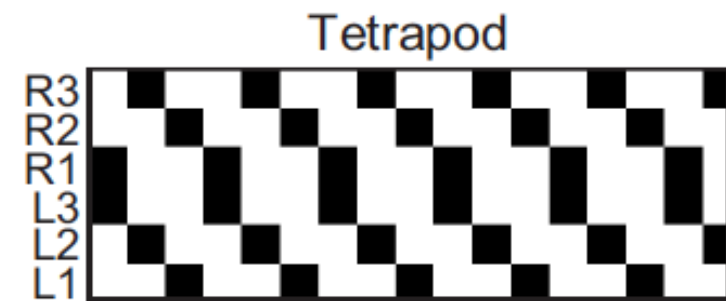
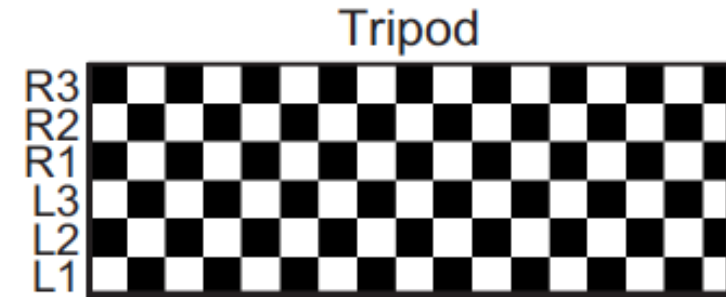
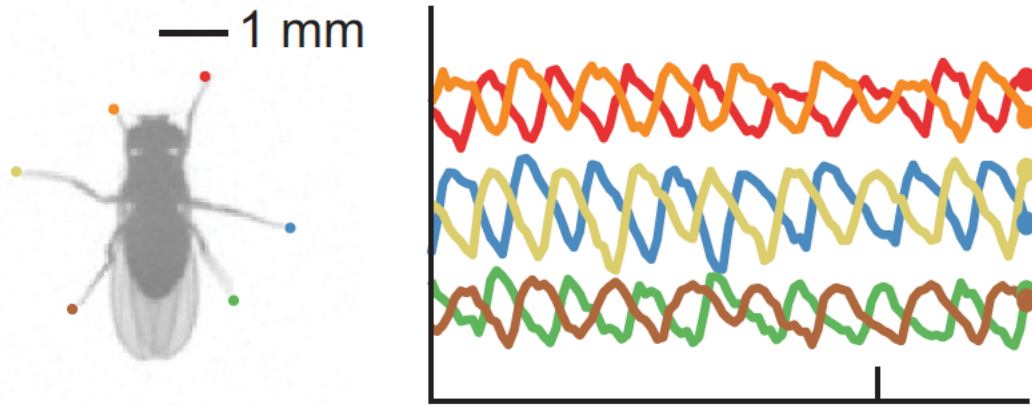
Barcelona Summer School for Advanced Modeling of Behavior,
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Canonical hexapod gaits



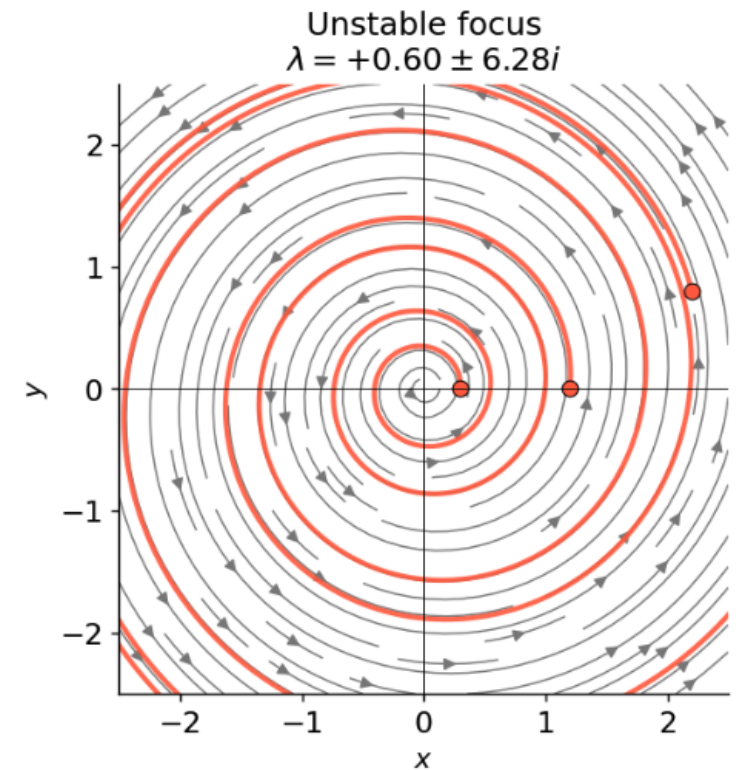
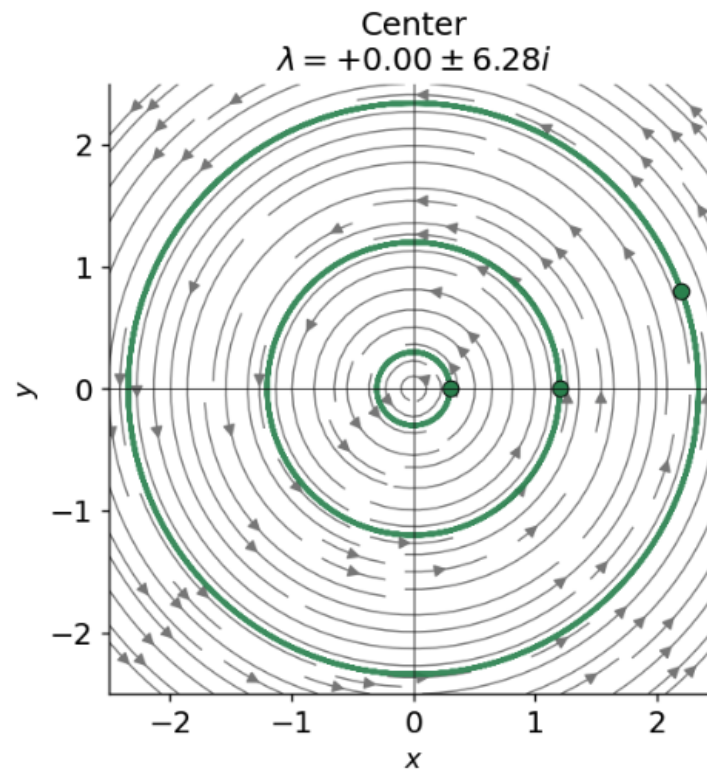
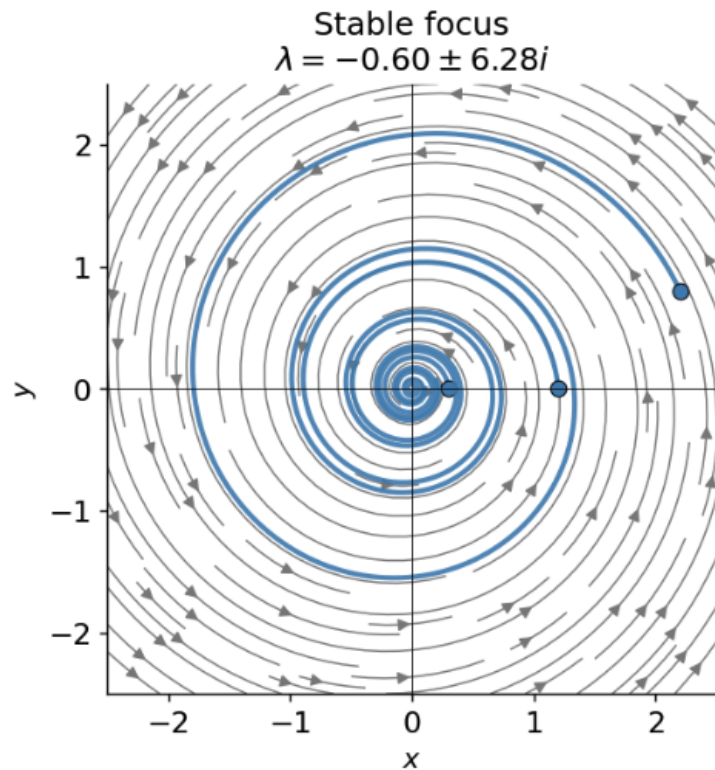
Outline

- Why do we need nonlinear dynamics to model fly walking?
- Kuramoto model - simplest possible model of leg coordination
- Stuart-Landau model (Hopf normal form)
- Perturbation and gait recovery as experimental test

Possible oscillatory behavior of linear dynamical systems

$$\dot{z} = Az, \quad A(\sigma, \omega) = \begin{pmatrix} \sigma & -\omega \\ \omega & \sigma \end{pmatrix}$$

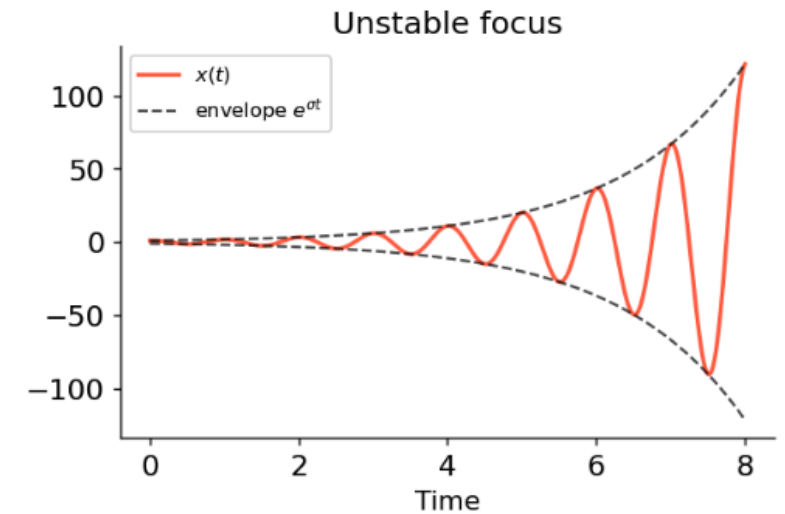
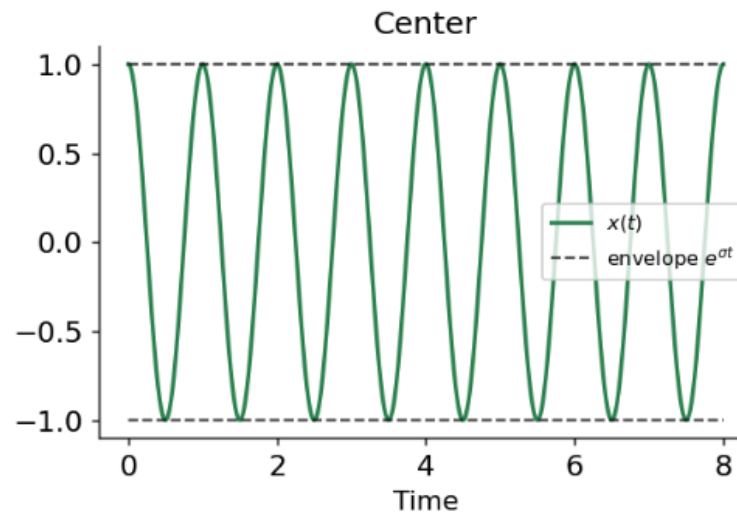
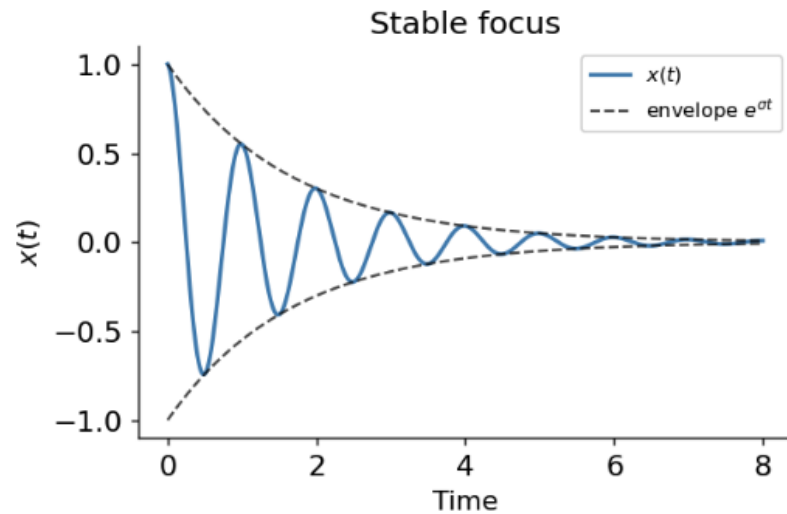
The **eigenvalues** of this system are: $\lambda = \sigma \pm i\omega$



Possible oscillatory behavior of linear dynamical systems

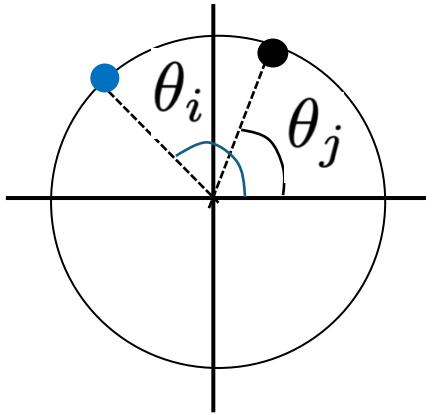
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Kuramoto model

Phase oscillators



$$\frac{d\theta_i}{dt} = \omega_i + \frac{1}{N} \sum_{j=1}^N \kappa_{ij} \sin(\theta_j - \theta_i)$$

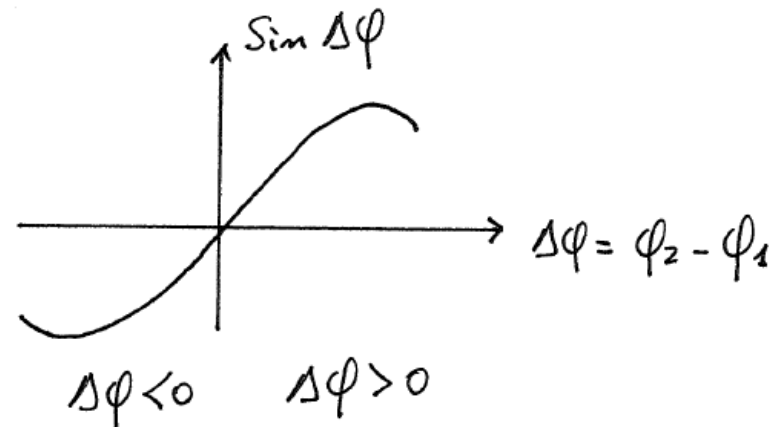
Single node (single fly leg):

$$\frac{d\theta_i}{dt} = \omega_i = \frac{2\pi}{T} = 2\pi f$$

$N = 2$ (left / right leg):

$$\frac{d\theta_1}{dt} = \omega_1 + \kappa \sin(\theta_2 - \theta_1)$$

$$\frac{d\theta_2}{dt} = \omega_2 + \kappa \sin(\theta_1 - \theta_2)$$

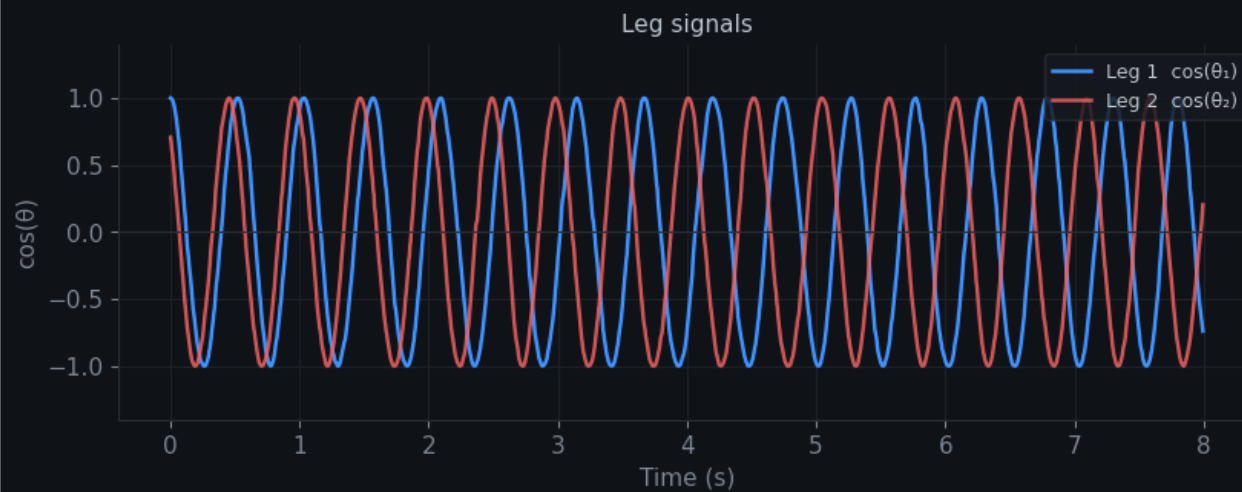
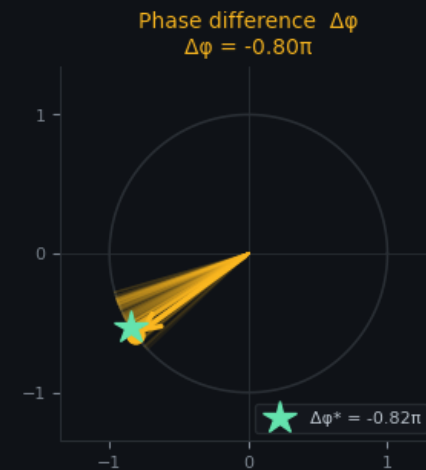
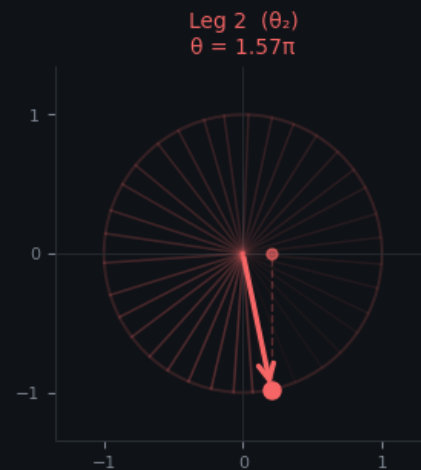
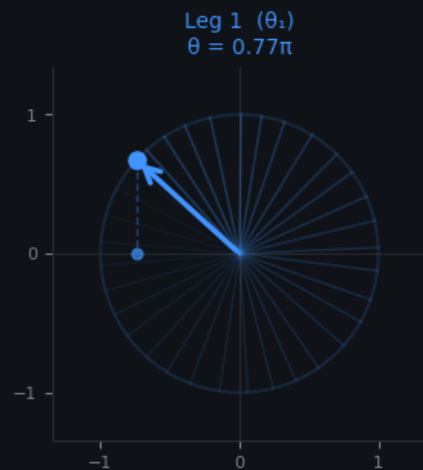


f_1 (Hz) 2.00
 f_2 (Hz) 1.90
 κ (coupling) -0.60

σ (noise) 0.20
 $\theta_1(0) / \pi$ 0.00
 $\theta_2(0) / \pi$ 0.25

► Run simulation

Phase-locked (partial lock) $\Delta\phi^* = -0.82\pi$ $|\Delta\omega| = 0.63 < 2|\kappa| = 1.20$ rad/s



Stable fixed points of the 2-oscillator Kuramoto model

$N = 2$ (left / right leg):

$$\begin{aligned}\frac{d\theta_L}{dt} &= \omega_L + \kappa \sin(\theta_R - \theta_L) \\ \frac{d\theta_R}{dt} &= \omega_R + \kappa \sin(\theta_L - \theta_R)\end{aligned}$$

Subtract the two phase equations:

$$\begin{aligned}\Delta\phi &= \theta_L - \theta_R \\ \Delta\omega &= \omega_L - \omega_R \\ \dot{\Delta\phi} &= \Delta\omega + \kappa \sin(-\Delta\phi) - \kappa \sin(\Delta\phi) = \Delta\omega - 2\kappa \sin(\Delta\phi)\end{aligned}$$

Fixed points. Set $\dot{\Delta\phi} = 0$: $\sin(\Delta\phi^*) = \frac{\Delta\omega}{2\kappa}$

Thus, we must have: $\boxed{|\Delta\omega| \leq 2|\kappa|}$ **locking condition**

Two solutions: $\Delta\phi_A^* = \arcsin\left(\frac{\Delta\omega}{2\kappa}\right)$ stable for $\kappa > 0$

$$\Delta\phi_B^* = \pi - \arcsin\left(\frac{\Delta\omega}{2\kappa}\right) \text{ stable for } \kappa < 0$$

For equal frequencies, we get

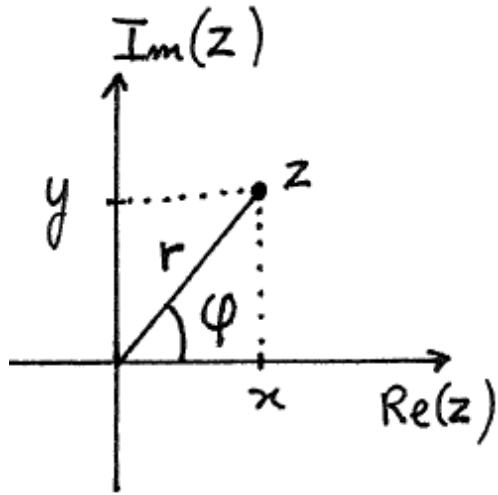
• $\Delta\phi_A^* = 0$ stable in-phase

• $\Delta\phi_B^* = \pi$ stable anti-phase

Limitations of the Kuramoto model

- **No concept of amplitude.** It cannot represent the *onset* and *offset* of walking (amplitude rising from 0 to r^*)
- It thus cannot explain the **effect of amplitude perturbations** (we will see later that a striking experimental feature is amplitude disruption and recovery - invisible to Kuramoto)
- **Phase is not directly observable.** You observe leg position $x_k(t)$, not phase $\theta_k(t)$. The waveform is highly non-sinusoidal (e.g., stance is much longer than swing - a common feature of slow walking)

Stuart-Landau / Hopf Normal Form



$$z = x + iy$$

$$z = r e^{i\varphi} = r \cos \varphi + i r \sin \varphi$$

$$\varphi = \arctan\left(\frac{y}{x}\right)$$

$$|z| = r = \sqrt{x^2 + y^2}$$

Single oscillator (single leg) :

$$\dot{z} = (\mu + i\omega)z - |z|^2 z$$

In polar coordinates:

$$\dot{r} = r(\mu - r^2)$$

$$\dot{\theta} = \omega$$

constant rotation at ω ,
independent of amplitude

The Hopf bifurcation

Radial equation:

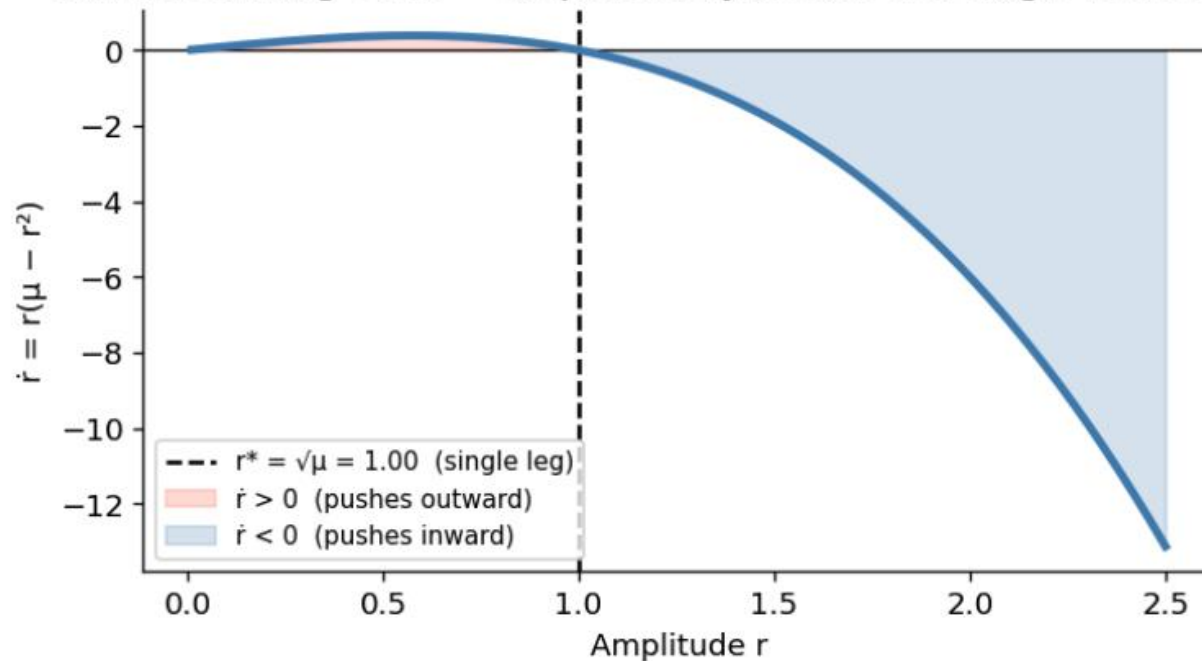
$$\dot{r} = r(\mu - r^2)$$

Fixed points:

- $r = 0$: stable when $\mu < 0$, unstable when $\mu > 0$
- $r^* = \sqrt{\mu}$: exists and is stable when $\mu > 0$

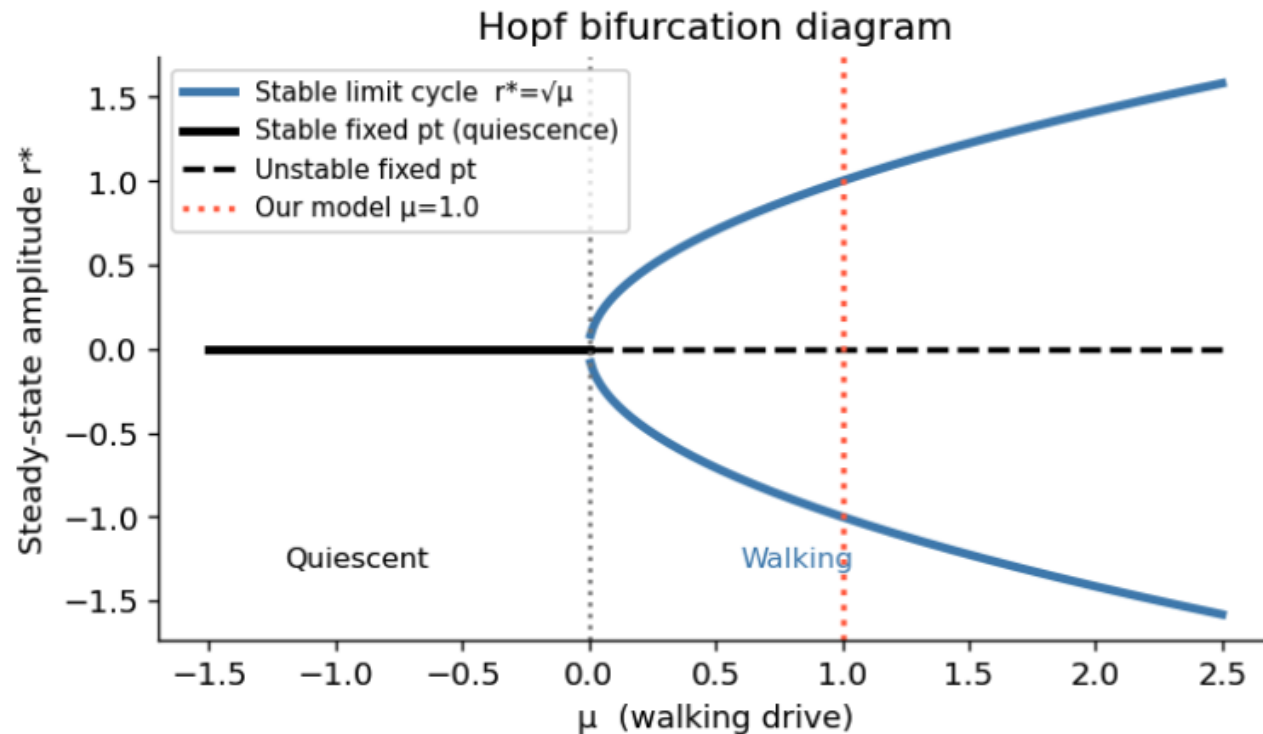
$$\mu = 1$$

Radial restoring force — amplitude dynamics of a single SL oscillator

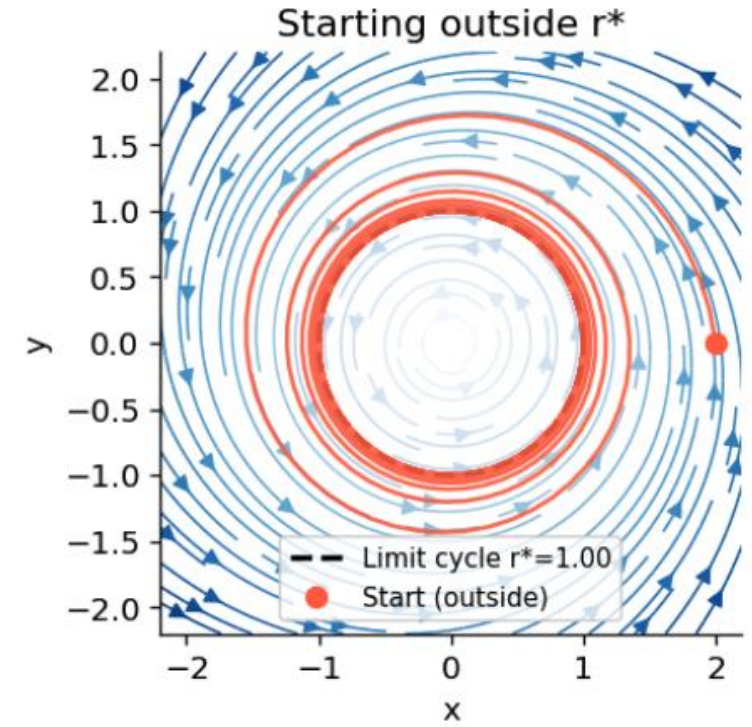
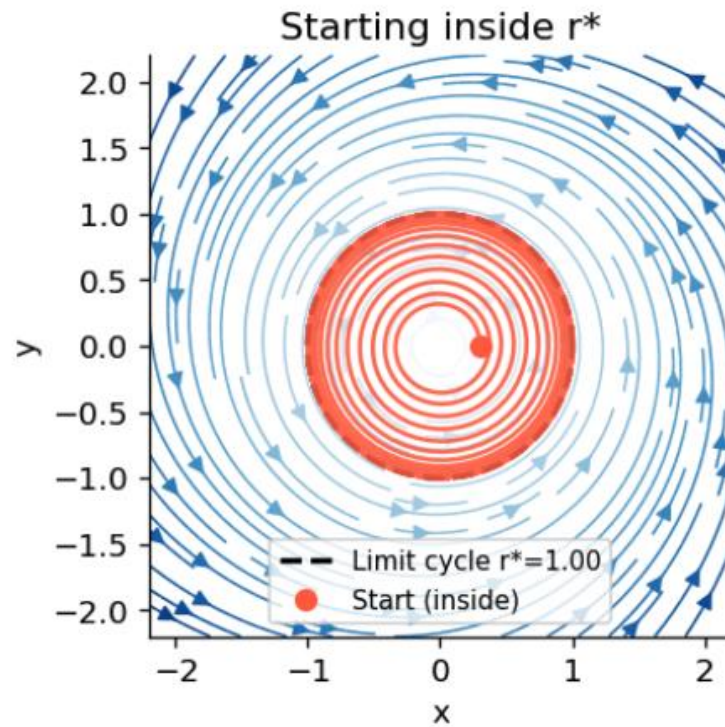
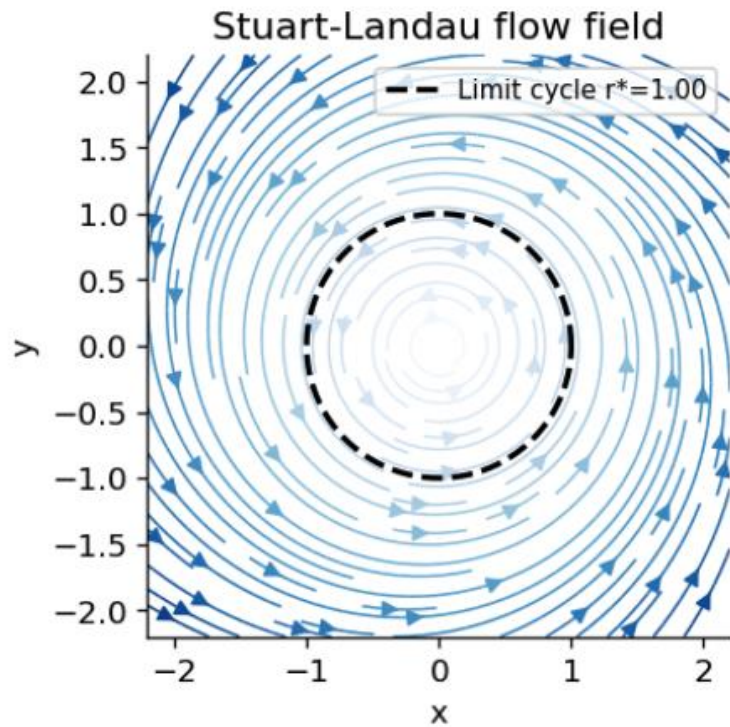


The Hopf bifurcation

- $\mu < 0$: stable fixed point at $z = 0$ (quiescence)
- $\mu = 0$: bifurcation point
- $\mu > 0$: stable limit cycle at $r^* = \sqrt{\mu}$ (oscillation)

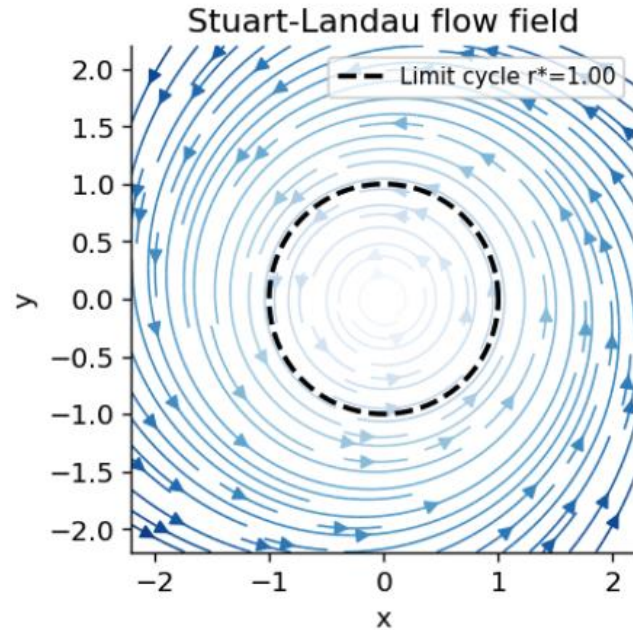


Limit cycle – flow field



Trajectories are attracted to the limit cycle from everywhere!

Connection to Kuramoto



Two coupled Stuart-Landau oscillators (two legs) :

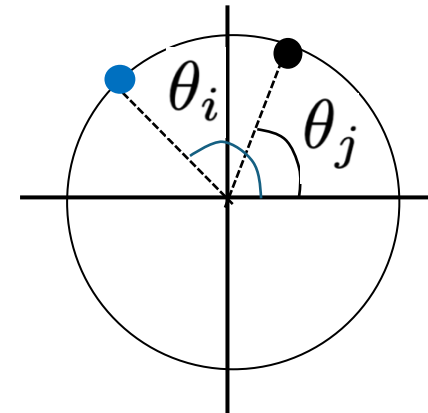
$$\dot{z}_k = (\mu + i\omega)z_k - |z_k|^2 z_k + \kappa(z_j - z_k)$$

Limit cycle (for two anti-phase oscillators):

$$r_{\text{coupled}}^* = \sqrt{\mu - 2\kappa}$$

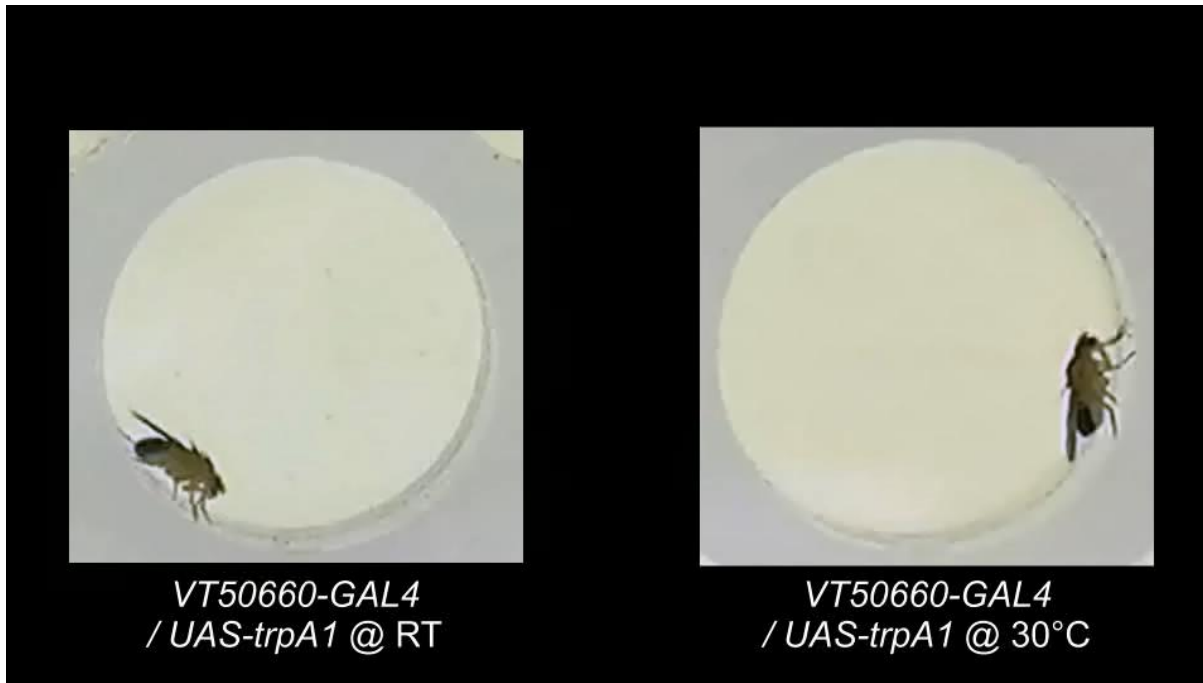
Near the limit cycle and assuming small noise, this reduces (to leading order) to the Kuramoto model:

$$\dot{\theta}_k = \omega + \kappa \sin(\theta_j - \theta_k)$$



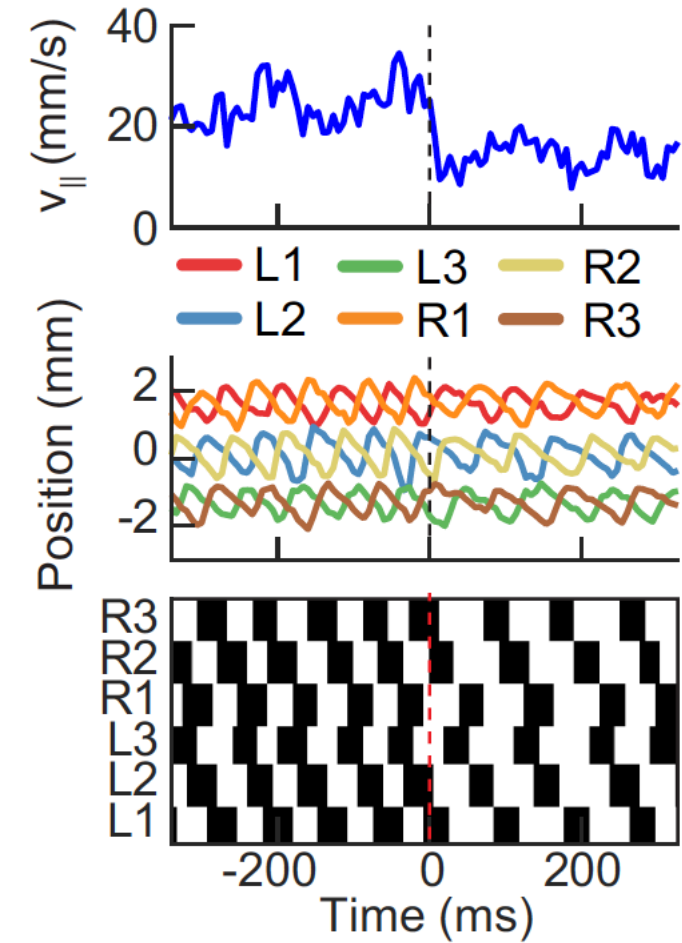
Perturbation experiments

“Moonwalker” neurons



Bidaye et al. 2014

Brief optogenetic stimulation



DeAngelis, Zavatone-Veth et al. 2019;
DeAngelis et al. 2020

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